

Quantum Dynamics

Schrödinger Dynamics:

$$|\psi, t\rangle = \sum_i |u_i\rangle \underbrace{c_i(t)}_{\text{time-dep.}}$$

Unitary time Evolution: $\mathcal{U}(t, t_0)$

$$|\psi, t\rangle = \mathcal{U}(t, t_0) |\psi, t_0\rangle \quad \forall t, t_0$$

$$\mathcal{U}^\dagger(t, t_0) \mathcal{U}(t, t_0) = \mathbb{1}$$

$$\hookrightarrow \text{Thus } \langle \psi, t | \psi, t \rangle = \langle \psi, t_0 | \psi, t_0 \rangle$$

$$\cdot \mathcal{U}(t_0, t_0) = \mathbb{1}$$

$$\cdot \mathcal{U}(t_2, t_0) = \mathcal{U}(t_2, t_1) \cdot \mathcal{U}(t_1, t_0)$$

$$* \cdot \mathcal{U}(t_0, t_1) = \mathcal{U}^\dagger(t_1, t_0)$$

Schrödinger's Equation.

$$i\hbar \partial_t |\psi, t\rangle = H(t) |\psi, t\rangle$$

$$H(t) = i\hbar \frac{\partial \mathcal{U}(t, t_0)}{\partial t} \cdot \mathcal{U}^\dagger(t, t_0)$$

Poisson's bracket

$$\hookrightarrow \{A, B\}_{pb} = \frac{\partial A}{\partial n} \frac{\partial B}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial B}{\partial n}$$

$$\bullet Q(n, p) \rightarrow \frac{dQ}{dt} = \{Q, H\}_{pb}$$

$$\bullet [A, B] \Rightarrow i\hbar \{A, B\}_{pb}$$

One-Dimensional Sch-Eq.

$$i\hbar \frac{\partial \psi(n, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial n^2} + V(n) \right) \psi(n, t)$$

$\Rightarrow$ Calculating $\mathcal{U}$:

$$\frac{d\mathcal{U}}{dt} = -\frac{i}{\hbar} H(t) \mathcal{U}(t)$$

1. Time-independent H .

$$\mathcal{U}(t, t_0) = \exp \left\{ -\frac{i}{\hbar} H(t - t_0) \right\}$$

2. Time-dep: $[H(t_1), H(t_2)] = 0$

$$U(t, t_0) = \exp \left\{ \frac{-i}{\hbar} \int_{t_0}^t dt' H(t') \right\}$$

3. Time-dep: $[H(t_1), H(t_2)] \neq 0$

$$U(t, t_0) = 1 + (-i/\hbar) \int_{t_0}^t dt_1 H(t_1) + (-i/\hbar)^2 \int_{t_0}^t dt_1 H(t_1) \int_{t_0}^{t_1} dt_2 H(t_2) + \dots$$

$\vdots$

$\underline{\underline{t_n}}$

Heisenberg Dynamics:

$\hat{A}_S$: Schrodinger operator

$$\langle \alpha, t | \hat{A}_S | \beta, t \rangle = \langle \alpha, 0 | U^\dagger(t, 0) \hat{A}_S U(t, 0) | \beta, 0 \rangle$$

$$\Rightarrow \boxed{\hat{A}_H(t) = U^\dagger(t, 0) \hat{A}_S U(t, 0)}$$

• $\hat{A}_H(0) = \hat{A}_S$

• If $\hat{C}_S = \hat{A}_S \hat{B}_S \Rightarrow \hat{C}_H = \hat{A}_H \hat{B}_H$

• $[\hat{A}_S, \hat{B}_S] = \hat{C}_S \Rightarrow [\hat{A}_H, \hat{B}_H] = \hat{C}_H$

• $\hat{H} = \hat{H} \quad \Rightarrow \quad \int_{t_1}^{t_2} dt \quad U^\dagger(t) \hat{H}(t) U(t) = \hat{H}$

$$I_H = I_S \quad \text{when} \quad (I_S(t), I_S(t)) = 0$$

$$\cdot \langle \psi, t | \hat{A}_S | \psi, t \rangle = \langle \psi, 0 | \hat{A}_H | \psi, 0 \rangle$$

$$\langle \hat{A}_S \rangle = \langle \hat{A}_H(t) \rangle$$

Heisenberg equation of Motion

$$i\hbar \frac{d\hat{A}_H(t)}{dt} = [\hat{A}_H(t), \hat{H}_H(t)] + i\hbar \left(\frac{\partial \hat{A}_S(t)}{\partial t} \right)_H$$

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Note :

1. $\hat{A}_S$ is time-indp.

$$\hookrightarrow \cdot i\hbar \frac{d\hat{A}_H(t)}{dt} = [\hat{A}_H(t), \hat{H}_H(t)]$$

$$L_2 \cdot i\hbar \frac{d}{dt} \langle \hat{A}_H(t) \rangle = \langle [\hat{A}_H(t), \hat{H}_H(t)] \rangle$$

↳ $[\hat{A}_s, H_s] = 0$

Heisenberg operators for SHO:

$$\hat{x}_H(t) = \hat{x} \cos \omega t + \frac{1}{m\omega} \hat{p} \sin \omega t$$

$$\hat{p}_H(t) = \hat{p} \cos \omega t - m\omega \hat{x} \sin \omega t$$

Time-Dependent $\hat{a}$ & $\hat{a}^\dagger$

$$\hat{a}(t) = e^{-i\omega t} \hat{a}$$

$$\hat{a}^\dagger(t) = e^{i\omega t} \hat{a}^\dagger$$

$\Rightarrow$ Translation:

$$T_{x_0} = e^{-\frac{i}{\hbar} \hat{p} x_0}$$

$$T_{x_0} |x_1\rangle = |x_1 + x_0\rangle$$

$$\langle x_1 | T_{x_0} = \langle x_1 - x_0 |$$

Uncertainty

$$\Delta \hat{A} \Delta \hat{B} \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

Coherent States:

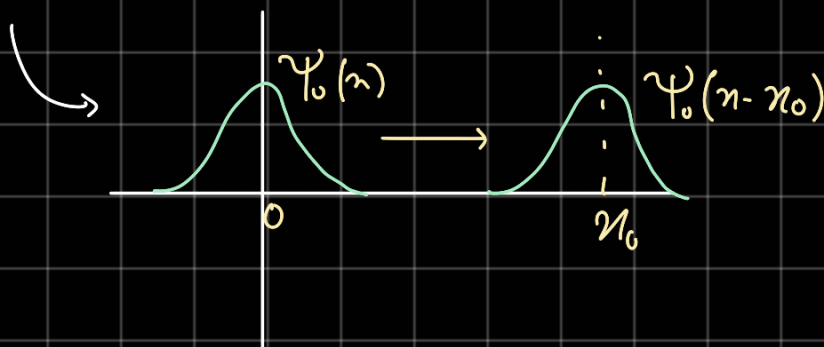
Defined as $\tilde{\chi}_0$: Not a position state.

$$|\tilde{\chi}_0\rangle \equiv T_{x_0}|0\rangle = e^{\frac{-i}{\hbar}\hat{p}x_0}|0\rangle$$

where, $|0\rangle \rightarrow$ zero state

Basically, Translation of ground state by a dist x_0

Wave function: $\psi_{x_0}(x) = \langle x|\tilde{\chi}_0\rangle = \langle x|T_{x_0}|0\rangle$
 $= \langle x-x_0|0\rangle = \psi_0(x-x_0)$



$\Rightarrow$ Expectation value of $\hat{n}$:

$$\begin{aligned}\langle \tilde{\chi}_0|\hat{n}|\tilde{\chi}_0\rangle &= \langle 0|T_{x_0}^\dagger \hat{n} T_{x_0}|0\rangle \\ &= \langle 0|(\hat{n}+x_0)|0\rangle \\ &= x_0\end{aligned}\quad \left[\begin{array}{l} T_{x_0}^\dagger \hat{n} T_{x_0} = \hat{n} + x_0 \\ \langle 0|\hat{n}|0\rangle = 0 \end{array} \right]$$

$\Rightarrow$ Expectation value of $\hat{p}$:

$$\begin{aligned}\langle \tilde{\chi}_0|\hat{p}|\tilde{\chi}_0\rangle &= \langle 0|T_{x_0}^\dagger \hat{p} T_{x_0}|0\rangle \\ &= \langle 0|\hat{p} T_{x_0}^\dagger T_{x_0}|0\rangle \\ &= \langle 0|\hat{p}|0\rangle \\ &= 0\end{aligned}\quad [\hat{p} \text{ commutes with } T_{x_0}]$$

$\Rightarrow$ Expectation value of Energy :

$$\langle \tilde{\chi}_0 | H | \tilde{\chi}_0 \rangle = \langle 0 | T_{\chi_0}^\dagger H T_{\chi_0} | 0 \rangle$$

$$T_{\chi_0}^\dagger H T_{\chi_0} = T_{\chi_0}^\dagger \left(\frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \right) T_{\chi_0}$$

Recall: $T_{\chi_0}^\dagger \hat{x} T_{\chi_0} = (\hat{x} + \chi_0)$

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 (\hat{x} + \chi_0)^2$$

$$= H + m \omega^2 \chi_0 \hat{x} + \frac{1}{2} m \omega^2 \chi_0^2$$

Thus,

$$\begin{aligned} \langle \tilde{\chi}_0 | H | \tilde{\chi}_0 \rangle &= \langle 0 | H | 0 \rangle + m \omega^2 \chi_0 \langle 0 | \hat{x} | 0 \rangle + \frac{1}{2} m \omega^2 \chi_0^2 \\ &= \frac{1}{2} \hbar \omega + 0 + \frac{1}{2} m \omega^2 \chi_0^2 \end{aligned}$$

$\Rightarrow$ We see that,

The coherent state is the quantum version of a point particle on a spring held stretched to $x = \chi_0$.

